Group Decis Negot (2012) 21:461–473
DOI 10.1007/s10726-010-9218-2

A Study on Aggregation of TOPSIS Ideal Solutions for Group Decision-Making

Yeu-Shiang Huang · Wei-Hao Li

Published online: 1 December 2010

Abstract The technique for order preference by similarity to ideal solution (TOPSIS) has become a popular multi-criteria decision making (MCDM) technique, since it has a comprehensible theoretical structure and is able to provide an exact model for decision making. For the use of TOPSIS in group decisions, the common approaches in aggregating individual decision makers' judgments are the geometric and the arithmetic mean methods, although these are too intuitive and do not consider either preference levels or preference priorities among alternatives for individual decision makers. In this paper, a TOPSIS group decision aggregation model is proposed in which the construction consists of three stages: (1) The weight differences are calculated first as the degrees of preferences among different alternatives for each decision maker; (2) The alternative priorities are then derived, and the highest one can be denoted as the degree to which a decision maker wants his most favorite alternative to be chosen; (3) The group ideal solutions approach in TOPSIS is used for the aggregation of similarities obtained from different decision makers. A comparative analysis is performed, and the proposed aggregation model seems to be more satisfactory than the traditional aggregation model for solving compromise-oriented decision problems.

Keywords Multi-criteria decision making · TOPSIS · Group decisions · Preference aggregation

Notations

n	the number of alternatives
p	the number of decision makers

Y.-S. Huang (✉) · W.-H. Li
Department of Industrial and Information Management, National Cheng Kung University,
1 University Rd., Tainan 701, Taiwan
e-mail: yshuang@mail.ncku.edu.tw

D	decision matrix
C_{ki}	relative closeness of alternative i by decision maker k
α_{kij}	weighted difference between alternatives i and j by decision maker k
α_k	weighted difference for decision maker k
φ_{ki}	preference ranking of alternative i by decision maker k
φ_{Gi}	preference ranking of alternative i by the group
β_{ki}	preference priorities of alternative i by decision maker k
$\beta_i^{(G)}$	preference priorities of alternative i by the group
G	group decision matrix
$C_{ki}^{(w)}$	revised relative closeness of alternative i by decision maker k
G'	weighted group decision matrix
v_{ki}	weighted relative closeness of alternative i by decision maker k
A_G^*	group positive ideal solution set
A_G^-	group negative ideal solution set
v_{Gk}^*	best scores with respect to decision maker k by the group
v_{Gk}^-	worst scores with respect to decision maker k by the group
S_{Gi}^*	separation of alternative i with respect to group positive ideal solution by the group
S_{Gi}^-	separation of alternative i with respect to group negative ideal solution by the group
C_{Gi}	relative closeness of alternative i by the group
η_{ki}	differences of alternative i between decision maker k and the group
η'_{ki}	normalized differences of alternative i between decision maker k and the
S_{Gi}^-	group
η_k	mean differentiation coefficient by decision maker k
ζ_{ki}	rank differences by alternative i between decision maker k and the group
a	smoothing constant
r_k	revised rank correlation by decision maker k
a_{ki}	weight by smoothing constant of alternative i by decision maker k
$r_k^{(d)}$	decision's rank correlation by decision maker k
τ_k	satisfaction index by decision maker k
τ	satisfaction index by the group

1 Introduction

The technique for order preference by similarity to ideal solution (TOPSIS), proposed by Hwang and Yoon (1981), has become a popular decision technique in the field of multi-criteria decision making (MCDM). However, the common approaches in aggregating individual decision makers' judgments for TOPSIS in group decision making are the geometric or the arithmetic mean methods, which are intuitive and do not consider either preference levels or preference priorities among alternatives for individual decision makers. For example, some decision makers in a group may have extreme preferences about some alternatives, whereas others may have similar preferences for all alternatives. In such a case, these mean approaches for group decision problems may result in a poor consensus decision and cause some implementation problems due to a lack of commitment. Accordingly, without considering the preferential degrees and priority ranking among alternatives for each decision maker, the traditional aggregation approaches of TOPSIS for group decision-making may produce a final decision which dissatisfies some group members and fails to inspire a collective commitment.

TOPSIS differs from other MCDM methods in its logical thinking, which is based on the simultaneous evaluation of the nearest distance from the best alternative (positive ideal solution; FPIS) and the longest distance from the worst alternative (negative ideal solution; FNIS). Shih et al. (2007) pointed out that TOPSIS is a straightforward technique and suitable for cases with a large number of attributes and alternatives, and especially appropriate for use with objective or quantitative data. Moreover, TOPSIS has many advantages: (i) a compromise can be obtained efficiently; (ii) it uses logical thinking that represents the rationale of human choice; (iii) it has a scalar value that expresses both the best and worst alternatives simultaneously; (iv) it uses a comprehensible computation process that can be easily programmed into a spreadsheet; (v) the performance measures of all alternatives on attributes can be visualized on a polyhedron, at least for any two dimensions; and (vi) the results are easily explained to and accepted by decision makers (Shih et al. 2007; Abo-Sinna and Amer 2005). In recent years, TOPSIS has been extensively applied in various fields of research and real-word problems, such as financial events (Deng et al. 2000), transportation (Janic 2003), facility location (Chu 2002), product design (Lin et al. 2008a,b), customer satisfaction (Lin and Chang 2008), personal decisions (Chen and Tzeng 2004), and service quality (Tsaur et al. 2002). This paper is motivated to focus on TOPSIS with intent to appropriately extend its underlying concepts to deal with the problems of group decision making.

Research on the criticisms of TOPSIS has been widely discussed in the literature. Firstly, TOPSIS does not provide a way for weight elicitation, although this problem is often resolved by using other techniques, such as the analytic hierarchy process (AHP) (Tsaur et al. 2002) or analytic network process (ANP) (Shyur and Shin 2006). Secondly, the normalization methods used in the decision matrix built by TOPSIS may influence the decision results. Shih et al. (2007) stated five normalization methods for decision makers, and gave the corresponding suitable conditions for each method. Finally, the measure of relative distances for ranking alternatives is still open to question, and in addition to Euclidean distance, different measures have been used, such as the least absolute value terms (Olson 2004), Minkowskis metrics (Lin et al. 2008a,b), and weighted Euclidean distance (Shyur and Shin 2006).

Originally, TOPSIS was designed for individual decision making (Hwang and Yoon 1981), but there are many preference aggregation approaches for TOPSIS and applications for group decision problems in the literature (Chen 2000; Chu 2002; Shih et al. 2004). With regard to group decision making, as mentioned previously, the common aggregation approaches are the geometric and the arithmetic mean methods, and they differ from each other in the ways they are utilized at each step in TOPSIS when aggregating group judgments, such as relative closeness (Parkan and Wu 1998), separation measure (Chu 2002), PIS and NIS (Shih et al. 2007), and fuzzy PIS and NIS (Chen 2000). In addition, research on the use of different decision techniques for preference aggregation in TOPSIS has also been conducted. Some researchers utilized the Borda count method to aggregate the preferences of decision makers (Pochampally and Gupta 2004; Shih et al. 2004, 2005). Fang and Zhou (2007) used the relative closeness of decision makers to construct a decision matrix, and employed it to calculate the group ideal solution which can aggregate the preferences of all individuals.

This study proposes a group decision method which is based on TOPSIS to include two indices of weight differences and alternative priorities, and which aggregates individual judgments by the group ideal solutions approach. The construction of the proposed model includes three stages: (1) The weight differences are calculated first as the degrees of preference among different alternatives for each decision maker; (2) The alternative priorities are then derived and the highest ranking can be denoted as the degree to which a decision maker wants his most favorite alternative to be chosen; (3) The group ideal solutions approach in TOPSIS is used for the aggregation of similarities obtained from different decision makers. This papers is organized as follows: Sect. 2 states the group decision problem; Sect. 3 describes the development of the proposed group TOPSIS approach; Sect. 4 shows that the proposed approach seems more satisfactory and effective than the traditional one for decision makers by performing a virtual case of group decision making; finally, Sect. 5 states the concluding remarks.

2 Group Decision Making in TOPSIS

Janic (2003) stated that the TOPSIS method embraces seven sub-steps which are as follows: (1) constructing the normalized decision matrix by using the decision making matrix; (2) constructing the weighted-normalized decision matrix; (3) determining the ideal and negative ideal solution; (4) calculating the separation measure of each alternative from the ideal one; (5) calculating the relative distance of each alternative to the ideal and negative ideal solution; (6) ranking the alternatives in descending order with respect to relative distance to the ideal solution; (7) identification of the preferable alternative as the closest to the ideal solution. However, in considering group decision making problems, the preferences among alternatives have to be aggregated for individual decision makers. After ranking the alternatives by utilizing TOPSIS individually, the geometric and the arithmetic mean methods are often used to aggregate the separation measures, ideal solutions or relative distances from individual decision makers. Usually, the mean approaches are suitable under a condition in which the decision makers are of equal importance and a consensus is about to be reached. In a real-world situation, some decision makers may strongly prefer some particular alternatives, and in the meantime, some other decision makers may prefer none at all. In such a case, the mean aggregation may result in a dissatisfactory final decision for some decision makers. Huang et al. (2009) states that an individual with greater preferential differences among alternatives would have more influence in a group than those who with less preferential differences, since in such a case, the individual with greater preferential differences would fight for his/her choices, while the other members may be less insistent because of their similar perceptions of all alternatives. The mean approaches may thus not be able to achieve a robust consensus, and a satisfactory level of commitment is not anticipated. Huang et al. (2009) proposed two indices, preferential differences and preferential ranks, to deal with this problem. However, their research was designed for AHP, and they integrated these two indices by an additive approach. This paper utilizes these two indices to the use of TOPSIS, and employs the logical thinking of TOPSIS to construct the proposed group decision making approach. TOPSIS logical thinking considers that the optimal decision should have the closest distance from the best alternative and the farthest distances from the worst alternative. In attempting to extend this thinking to the aggregation of individual decision makers' preferences, while the resulting choice may not be the one which is preferred by the group members in the first place, it can be the most satisfactory alternative, because it takes the worst ones into account. This preferences aggregation approach can thus avoid the problems which may occur using the common mean aggregation approaches.

Some extra factors may affect the process of preferences aggregation in practice, such as the risk attitudes of decision makers, or situations when other decision makers are not able to present their preferences due to the existence of a dominant individual. Therefore, there are some assumptions that have to be made to deal with the group decision problem: (a) All the decision makers in the group are able to state their preferences and make comparisons accordingly. (b) All the decision makers in the group are honest, and will not purposely overrate or underrate some alternatives. (c) The decision makers have the same power (weight), and a monopoly does not exist.

3 The Proposed Group TOPSIS Decision Model

Suppose a group decision problem has p decision makers and n alternatives. Each decision maker first evaluates and ranks the n alternatives individually by the original TOPSIS. After the ranking process, all decision makers can have their relative closeness of the n alternatives, and the aggregation procedure can thus be undertaken.

3.1 Preferential Differences Integration

Huang et al. (2009) pointed out that each decision maker may have his/her own preferential differences among alternatives, and they proposed an index to indicate this. In addition, Matsatsinis et al. (2005) also proposed another index for the preferential differences. Similarly, in this study, the weighted difference between alternatives i and j by decision maker k can be defined as

$$\alpha_{kij} = \left| C_{ki} - C_{kj} \right|, \quad 0 \leqq \alpha_{kij} \leqq 1, \quad i \neq j, \tag{3.1}$$

where C_{ki} is the relative closeness of alternative i evaluated by decision maker k. The mean approach is utilized to integrate the weighted differences for every pair of alternatives in this study. Note that α_{kij} only denotes the preferential differences for decision maker k, so we normalize the mean differences for the p decision makers to obtain the relative preferential differences in the group. Therefore, the weighted difference for decision maker k is given by

$$\alpha_k = \frac{\sum\limits_{i<j} \alpha_{kij}}{\sum\limits_{k=1}^{p} \sum\limits_{i<j} \alpha_{kij}}, \quad 0 < \alpha_k < 1, \quad i \neq j, \quad 1 \leqq i < j \leqq n, \tag{3.2}$$

where α_k is calculated by summing up α_{kij} for $\binom{n}{2}$ times in order to take all possible combinations of alternative pairs into account.

If α_k is a small value, the preferential difference for decision maker k is also small, which means he/she is unable to or has no interest in decisively discriminating these alternatives, and thus may have similar preferences for all of them. This can influence the determination of the final group decision. For example, suppose there are four decision makers, A, B, C and D, and the values of α_k for A, B, and C are small, but for D it is large. If the ranking of the aggregation results is not similar to D's preferences, D will be dissatisfied. However, if the ranking of aggregation results is not similar to the preferences of A, B, and C, they will not be as dissatisfied as D in the first case. Therefore, we utilize the weighted differences as a way for an individual decision maker to emphasize his/her preferential differences among the group.

3.2 Preferential Priorities Integration

Huang et al. (2009) also stated that decision makers would particularly care about whether their best ranked alternative is adopted in a group decision problem, and give more importance to this than to other alternatives. Based on this perspective, we define the preference priority of alternative i by decision maker k as $\beta_{ki} = n/\varphi_{ki}$, where φ_{ki} denotes the preference ranking of alternative i by decision maker k. Note that if some alternatives are tied with the same rank, they are all assigned the rank which is the average of their original ranking positions. For example, if two alternatives are tied with the same rank 2 which means they are in positions 2 and 3, then the new rank assigned to both of them would be 2.5. The use of the preference priority is able to emphasizes the importance of the best ranked alternative which is often much more important than other alternatives. Moreover, since all decision makers are assumed to have the same power (weight), the preference priorities of alternative i by all decision makers are aggregated as $\beta_i = \sum_{k=1}^{p} \beta_{ki}$. Accordingly, we normalize β_i to get the relative important degree of alternative i for the decision group, and the preference priority of alternative i by the group, $\beta_i^{(G)}$, can thus be derived as

$$\beta_i^{(G)} = \frac{\beta_i}{\sum\limits_{i=1}^{n} \beta_i}. \tag{3.3}$$

The preference priority particularly gives more weight to the best ranked alternative, and thus is realistic and pragmatic for decision makers because it is able to indicate the extent to which decision makers expect their most favorite alternative to be adopted in a group decision problem. Accordingly, it can be utilized to revise the relative closeness of decision makers, i.e., $C_{ki}^{(w)} = C_{ki}\beta_i^{(G)}$.

3.3 Integration from TOPSIS Thinking

The logical thinking of TOPSIS, which is based on the concept that the optimal decision has the nearest distance from the best alternative and the longest distances from the worst alternative, can be utilized to aggregate the preferences of decision makers. In such a case, a decision maker can be regarded as an attribute. Upon constructing a new group decision matrix which can be used to calculate the group ideal solutions and relative closeness, the preference aggregation can thus be achieved.

We define group decision matrix G as follows:

$$G = \begin{bmatrix} C_{11}^{(w)} & C_{12}^{(w)} & \cdots & C_{1n}^{(w)} \\ C_{21}^{(w)} & C_{22}^{(w)} & \cdots & C_{2n}^{(w)} \\ \vdots & \vdots & \ddots & \vdots \\ C_{p1}^{(w)} & C_{p2}^{(w)} & \cdots & C_{pn}^{(w)} \end{bmatrix}. \tag{3.4}$$

The matrix G is an un-weighted matrix, and by using α_k as weights, the weighted group decision matrix G' can be given by

$$G' = \begin{bmatrix} v_{11} & v_{21} & \cdots & v_{1n} \\ v_{21} & v_{22} & \cdots & v_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ v_{p1} & v_{p2} & \cdots & v_{pn} \end{bmatrix}, \tag{3.5}$$

where $v_{ki} = \alpha_k C_{ki}^{(w)}$ is the weighted relative closeness of alternative i evaluated by decision maker k. Note that v_{ki} is revised by using both preferential differences and preferential priorities, so G' is adjusted by the preferential conditions of decision makers. According to the weighted matrix, the group ideal solutions would be more realistic for decision makers, and the group positive and negative ideal solutions are defined as follows:

$$A_G^* = \left\{ v_{G1}^*, v_{G2}^*, \ldots, v_{Gk}^*, \ldots, v_{Gp}^* \right\}, \quad \text{and} \tag{3.6}$$

$$A_G^- = \left\{ v_{G1}^-, v_{G2}^-, \ldots, v_{Gk}^-, \ldots, v_{Gp}^- \right\}, \tag{3.7}$$

respectively, where $v_{Gk}^{*} = \left\{ \max_{i} v_{ki} \,|\, i = 1, 2, \ldots, n \right\}$ denotes the best scores with respect to decision maker k by the group and $v_{Gk}^{-} = \left\{ \min_{i} v_{ki} \,|\, i = 1, 2, \ldots, n \right\}$ denotes the worst scores with respect to decision maker k by the group.

According to Eqs. (3.6) and (3.7), the distances of alternative i to the group positive ideal solution $\left(S_{Gi}^{*}\right)$ and the group negative ideal solution $\left(S_{Gi}^{-}\right)$ can be calculated as $S_{Gi}^{*} = \sqrt{\sum_{k=1}^{p} \left(v_{ki} - v_{Gk}^{*}\right)^2}$ and $S_{Gi}^{-} = \sqrt{\sum_{k=1}^{p} \left(v_{ki} - v_{Gk}^{-}\right)^2}$, respectively. The relative closeness of alternative i by the group, C_{Gi}, can thus be derived as

$$C_{Gi} = \frac{S_{Gi}^{-}}{S_{Gi}^{*} + S_{Gi}^{-}}, \quad i = 1, 2, \ldots, n, \tag{3.8}$$

and which can be used to rank the alternatives and thus achieve an optimal group decision.

3.4 The Satisfactory Index

Huang et al. (2009) proposed a satisfactory index to elicit the decision makers' satisfaction about a specific group decision. In this paper, with additional consideration of the differentiation coefficient and rank correlation, a revised satisfactory index is proposed to better manifest the contentment of a group.

The differences between the relative closeness of an alternative aggregated by the group and the relative closeness of an alternative assessed by an individual decision maker, i.e. $\eta_{ki} = |C_{ki} - C_{Gi}|$, would be an essential aspect to consider to express a decision maker's satisfaction. The greater the difference is, the less satisfied the decision maker would be. Therefore, the normalized differentiation coefficient is defined as follows for the alternative i and for the decision maker k

$$\eta'_{ki} = \frac{\frac{1}{\eta_{ki}}}{\sqrt{\sum_{i=1}^{n} \left(\frac{1}{\eta_{ki}}\right)^2}} \tag{3.9}$$

Moreover, by utilizing the arithmetic mean to integrate the entire set of alternatives, the mean closeness differentiation coefficient of decision maker k can be obtained as $\eta_k = \sum_{i=1}^{n} \eta'_{ki}/n$.

In addition to the differences of the relative closeness of alternatives between the group and its individual decision makers, the differences about the ranking of alternatives are also important for decision makers, and these are given by $\zeta_{ki} = |\phi_{ki} - \phi_{Gi}|$, where φ_{ki} is the final ranking of alternative i evaluated by decision maker k, and φ_{Gi} is the final ranking of alternative i aggregated by the decision group. Note that the similar approach is used in dealing with the condition when a tie exists among some alternatives for both ϕ_{ki} and ϕ_{Gi} as in previous section. However, the ranking differentiation coefficient ζ_{ki} may not be sufficient in revealing to what degree a decision maker would be disappointed when the final ranking of alternatives is very different from his/her own. Therefore, by considering the ranking differentiation coefficient and the correlation for ordinal scales, the revised rank correlation coefficient r_k is defined as

$$r_k = 1 - \frac{6\sum_{i=1}^{n} a_{ki}\zeta_{ki}^2}{n(n^2-1)}, \quad -1 \le r_k \le 1, \tag{3.10}$$

where a_{ki} is a smoothing constant that denotes the weight of alternative i for decision maker k. Exponential smoothing is employed to ensure that a decision maker gives the biggest weight to his/her most favorite alternative. The smoothing constant is decided by the decision maker, and is based on the preferential degree of the most favorite alternative. In order to specify the relative difference of rank correlation between a decision maker and the decision maker with the highest rank correlation, $r_k^{(d)}$ is defined as $r_k^{(d)} = \left|r_k - max(r_1, r_2, \ldots, r_k, \ldots, r_p)\right|$ which indicates that the smaller $r_k^{(d)}$ is (approaching zero), the more satisfactory decision maker k would relatively be.

The satisfactory index for an individual decision maker k can be attained by integrating the differentiation coefficient and the decision's rank correlation. Since a greater value of η_k denotes more satisfactory for decision maker k and both $r_k^{(d)}$ and η_k are bounded from zero to one, the satisfactory index for decision maker k can be defined as $\tau_k = \eta_k^{r_k^{(d)}}$ based on the similar argument in Huang et al. (2009), and its value is also bounded from zero and one. Finally, by utilizing the arithmetic mean approach, the overall satisfactory index for the decision group is thus given by $\tau = \sum_{k=1}^{p} \tau_k/p$. Note that the greater the τ is, the more satisfactory the decision group would overall be.

4 Illustrative Example

To illustrate the usefulness of the proposed group decision model and evaluate the satisfaction degree of the resulting decision, a company purchasing decision problem is utilized. Suppose a company needs to purchase a batch of computers to upgrade its equipment. The decision group is composed of six decision makers with the same decision power. After some meetings and data collection, they select the important evaluating attributes and decide on seven candidates of computer retailers to choose from. The six decision makers first evaluate the candidates via TOPSIS individually, and the results of this evaluation are shown in Table 1.

Based on Table 1, the proposed group decision approach and the traditional group decision approach which aggregates the relative closeness of individual decision makers by the arithmetic mean are respectively used to obtain the final group decision, and the results are shown in Table 2.

As shown in Table 2, Alternative E will be adopted if the decision group utilizes the traditional group decision approach, while Alternative B will be adopted if the decision group utilizes the proposed group decision approach. Further investigation is performed to study the preferential differences among the decision makers. The weight differences for the decision makers are shown in Table 3.

According to Table 3, we can infer that decision makers 1 and 6 may have strong preferences for alternatives and can discriminate from them more decisively. They thus seem more likely to get satisfied or dissatisfied with the results of the group decision than the other decision makers. Moreover, the preference priorities for the decision group are shown in Table 4.

Note that the decision group may prefer Alternatives B and E to others, since the group preference priorities for these are the highest two. In addition, we can observe that the number of decision makers who prefer Alternative B the most is the highest in the decision group.

Based on the preferential differences, decision makers 1 and 6 have higher weight differences, and decision makers 2, 3, 4 and 5 have lower weight differences. If the decision is made via the traditional group decision approach, the result is unlikely to satisfy the group, and especially not decision makers 1 and 6, and thus the company would have difficulty in executing this decision. However, if the group decision is made via the proposed group decision approach, the evaluating results of the group would be more consistent, since decision makers 2, 3, 4 and 5 would not be very dissatisfied with the final results because of their lower weight differences.

Table 1 Evaluation results for the seven alternatives

Alternative	Decision maker					
	1	2	3	4	5	6
A	0.553	0.480	0.328	0.381	0.435	0.399
B	0.682	0.368	0.438	0.340	0.390	0.702
C	0.356	0.394	0.310	0.359	0.535	0.410
D	0.543	0.335	0.317	0.498	0.388	0.298
E	0.530	0.490	0.435	0.490	0.530	0.460
F	0.401	0.379	0.441	0.368	0.428	0.402
G	0.376	0.423	0.422	0.429	0.463	0.475

Table 2 Aggregation result of the group decision problem

Alternative	Traditional method		Proposed approach	
	C_{Gi}	Ranking	C_{Gi}	Ranking
A	0.4293	4	0.1823	3
B	0.4867	2	0.8562	1
C	0.3940	7	0.1229	7
D	0.3965	6	0.1664	4
E	0.4892	1	0.6107	2
F	0.4032	5	0.1363	6
G	0.4313	3	0.1650	5

Table 3 Weight differences for the decision makers

Decision maker	α_k
1	0.2488
2	0.1237
3	0.1253
4	0.1345
5	0.1276
6	0.2400

Table 4 Preference priorities for the decision group

Alternative	$\beta_i^{(G)}$
A	0.1200
B	0.1913
C	0.1255
D	0.1240
E	0.1875
F	0.1286
G	0.1232

Table 5 Overall satisfactory indexes and their CV

Smoothing constant	Traditional method		Proposed approach	
	τ	CV	τ	CV
0.9	0.7744	0.3131	0.8331	0.2460
0.8	0.7952	0.2778	0.8457	0.2198
0.7	0.8167	0.2409	0.8589	0.1933
0.6	0.8397	0.2031	0.8722	0.1664
0.5	0.8650	0.1657	0.8846	0.1394

Therefore, the proposed model is more suitable for the decision group, if achieving greater commitment is the major concern.

The group preference priorities shown in Table 4 show that Alternatives B and E are preferred by the decision group, and this is matched by the evaluation results of the traditional and the proposed approaches. However, the proposed approach is more consistent than the traditional one, since the preference priority of alternatives B is higher than that of other alternatives. The resulting alternative obtained by the proposed approach integrated by TOPSIS thinking aggregation may not be an alternative which is preferred by every group member, but it is the most satisfactory choice because the worst alternatives have been considered into the construction of the group TOPSIS decision model.

The overall satisfactory indexes for the two approaches for different smoothing constants are shown in Table 5. In addition, we calculate the coefficient of variation

(CV), i.e., the ratio of the standard deviation to the mean, of satisfactory indexes for each decision maker to measure the dispersion of satisfaction for them. Note that the smoothing constants are set to be not less than 0.5 to prevent the unreasonable condition in which the better ranked alternatives have fewer weights than the worse ranked ones, and the greater the smoothing constant is, the more weight would be assigned to the best ranked alternative.

It can be seen in Table 5 that the proposed approach is more satisfactory than the traditional one at all levels of smoothing constants. Furthermore, by comparing the CV values, the proposed approach gives more alike satisfactory indexes among the decision makers than that for the traditional one. This implies that, compared to the proposed approach, the traditional approach may provide some decision makers with very high degrees of satisfaction and, in the meantime, provide some other decision makers with very low degrees of satisfaction, which would cause some problems of further implementations. The two results show that the proposed approach seems more effective and satisfactory for the decision makers than the traditional one. Finally, note that the greater the smoothing constant, the lower the satisfactory index. This may indicate that the original weight distribution of the decision makers is close to the exponentially smoothed distribution when the smoothing constant is set at 0.5, which means that the decision makers are relatively conservative in assigning weight to their best ranked alternative, and the differences among such a weight and the weights of other alternatives with lower ranks are relatively small compared to that for greater smoothing constants. Moreover, as the smoothing constant approaching 0.9, the weight distribution is far too extreme, and thus may be very dissimilar to the real weight distribution of the decision makers.

5 Conclusions

In an age of extensive competition among organizations, managers search for efficiency and excellence in solving decision problems. An effective group decision mechanism would enhance the quality of the group decision making process, and thus improve the performance of the organization. This paper proposes a group decision model, which differs from the traditional ones, based on TOPSIS aggregation, and considers the three aspects of weight differences, alternative priorities, and group ideal solutions to be taken into the construction. Therefore, the proposed model would result in a decision which is more realistic and acceptable for decision makers and, moreover, avoid the difficulty of attaining commitment which may be result from the traditional group decision approaches. Due to the development of e-democracy and information technology, decision makers are now able to conclude a group decision without a face-to-face meeting. In such cases, decision makers only need to rank alternatives using any MCDM technique individually, and then the aggregation of their preferences could be attained by the proposed model. As a result, and without much time consumption, the problems of aggregation of preferences can be solved and the managerial operations of the organization can be enhanced.

Note that the proposed model is based on the assumption that the decision makers will always honestly report their preferences. Moreover, some criticisms of TOP-

SIS have been made in recent years, such as problems in weight elicitation and distance measurement. Further study may thus utilize some existing revised TOPSIS approaches to extend the proposed model. In addition, instead of using weight differences and alternative priorities to describe the preferences of decision makers, it may be interesting to use other indicators, such as range or different moments, to construct a group decision approach.

References

Abo-Sinna MA, Amer AH (2005) Extensions of TOPSIS for multi-objective large-scale nonlinear programming problems. Appl Math Comput 162(1):243–256

Chen C-T (2000) Extensions of the TOPSIS for group decision-making under fuzzy environment. Fuzzy Sets Syst 114(1):1–9

Chen M-F, Tzeng G-H (2004) Combining grey relation and TOPSIS concepts for selecting an expatriate host country. Math Comput Modell 40:1473–1490

Chu TC (2002) Facility location selection using fuzzy TOPSIS under group decision. Int J Uncertain Fuzz Knowl Based Syst 10(6):687–701

Deng H, Yeh C-H, Willis RJ (2000) Inter-company comparison using modified TOPSIS with objective weights. Comput Oper Res 27(10):963–973

Fang W-G, Zhou H (2007) A multi-attribute group decision-making method approaching to group ideal solution. In: Proceeding of 2007 IEEE international conference on frey systems and intelligent services. Nanjing, China, pp 815–819

Huang Y-S, Liao J-T, Lin Z-L (2009) A study on aggregation of group decisions. Syst Res Behav Sci 26(4):445–454

Hwang CL, Yoon KP (1981) Multiple attribute decision making methods and applications: a state-of-the-art survey New York. Springer, Berlin

Janic M (2003) Multicriteria evaluation of high-speed rail, transrapid maglev and air passenger transport in Europe. Trans Plan Technol 26(6):491–512

Lin H-T, Chang W-L (2008) Order selection and pricing methods using flexible quantity and fuzzy approach for buyer evaluation. Eur J Oper Res 187(2):415–428

Lin M-C, Wang C-C, Chen M-S, Chang CA (2008a) Using AHP and TOPSIS approaches in customer-driven product design process. Comput Indus 59(1):17–31

Lin Y-H, Lee P-C, Chang TP, Ting H-I (2008b) Multi-attribute group decision making model under the condition of uncertain information. Automat Construct 17(6):792–797

Matsatsinis NF, Grigoroudis E, Samaras A (2005) Aggregation and disaggregation of preferences for collective decision-making. Group Decis Negot 14(3):217–232

Olson DL (2004) Comparison of weights in TOPSIS models. Math Comput Modell 40(7-8):721–727

Parkan C, Wu M-L (1998) Process selection with multiple objective and subjective attributes. Product Plan Control 9(2):189–200

Pochampally KK, Gupta SM (2004) A business-mapping approach to multi-criteria group selection of collection centers and recovery facilities. In: IEEE international symposium on electronics and the environment, pp 321–326

Shih H-S, Shyur H-J, Lee ES (2007) An extension of TOPSIS for group decision making. Math Comput Modell 45(7-8):801–813

Shih H-S, Wang C-H, Lee ES (2004) A multiattribute GDSS for aiding problem-solving. Math Comput Modell 39(11–12):1397–1412

Shih H-S, Huang L-C, Shyur H-J (2005) Recruitment and selection processes through an effective GDSS. Comput Math Appl 50(10–12):1543–1558

Shyur H-J, Shih H-S (2006) A hybrid MCDM model for strategic vendor selection. Math Comput Modell 44(7–8):749–761

Tsaur S-H, Chang T-Y, Yen C-H (2002) The evaluation of airline service quality by fuzzy MCDM. Tour Manage 23(2):107–115